EXP 2

AIM : To model a simplified hexagonal cellular network and study how cell radius affects coverage, estimated user capacity and handoff as a mobile cross a cell boundary.

Procedure:

1. Run the MATLAB script below and record the cell area, estimated cell count and total illustrative capacity for each radius.

2. Inspect the hexagonal layout and confirm that adjacent cells share edges without gaps in the idealized tiling.

3. Inspect the mobile-path plot and note the simulated handoff point.

4. Change the service area, radii or users-per-cell assumption one at a time; rerun and compare results.

5. Explain why the geometric cell count and nearest-center handoff differ from an operational cellular network.

OBJ -1:CODE

```
clear; clc; close all;

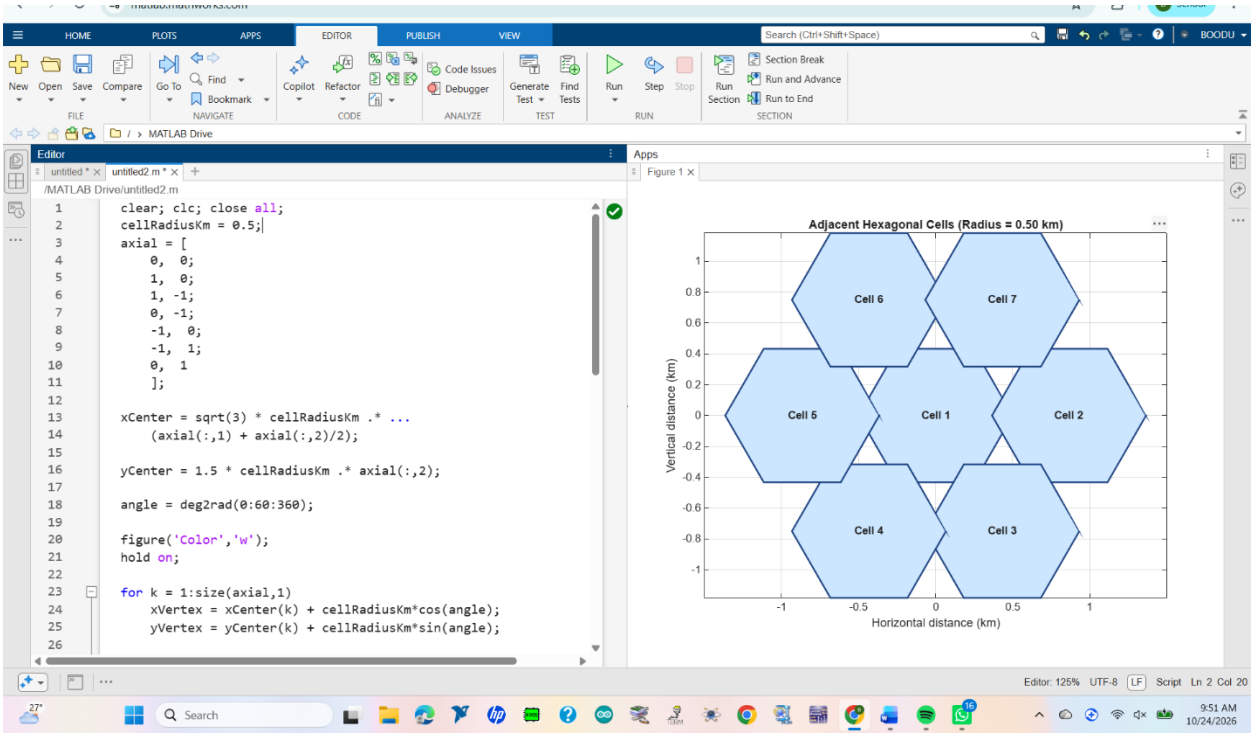
cellRadiusKm = 0.5;

axial = [
    0,  0;
    1,  0;
    1, -1;
    0, -1;
   -1,  0;
   -1,  1;
    0,  1
```

```
    ];
xCenter = sqrt(3) * cellRadiusKm .* ...
    (axial(:,1) + axial(:,2)/2);\
yCenter = 1.5 * cellRadiusKm .* axial(:,2);
angle = deg2rad(0:60:360);
figure('Color','w');
hold on;
for k = 1:size(axial,1)
    xVertex = xCenter(k) + cellRadiusKm*cos(angle);
    yVertex = yCenter(k) + cellRadiusKm*sin(angle);
    patch(xVertex, yVertex, [0.80 0.90 1.00], ...
        'EdgeColor',[0.10 0.25 0.50], ...
        'LineWidth',1.5);
    text(xCenter(k), yCenter(k), sprintf('Cell %d',k), ...
        'HorizontalAlignment','center', ...
        'FontWeight','bold');
end


axis equal;
grid on;
box on;
xlabel('Horizontal distance (km)');
ylabel('Vertical distance (km)');
title(sprintf('Adjacent Hexagonal Cells (Radius = %.2f km)', ...
    cellRadiusKm));
hold off;
```

OUTPUT:



OBJ-2:CODE

```
clear;

clc;

close all;

%% Changed Parameters

studyAreaKm2 = 6.0;

cellRadiiKm = [1.20, 0.90, 0.60];

%% Calculate Hexagonal Cell Area

cellAreaKm2 = (3 * sqrt(3) / 2) .* cellRadiiKm.^2;

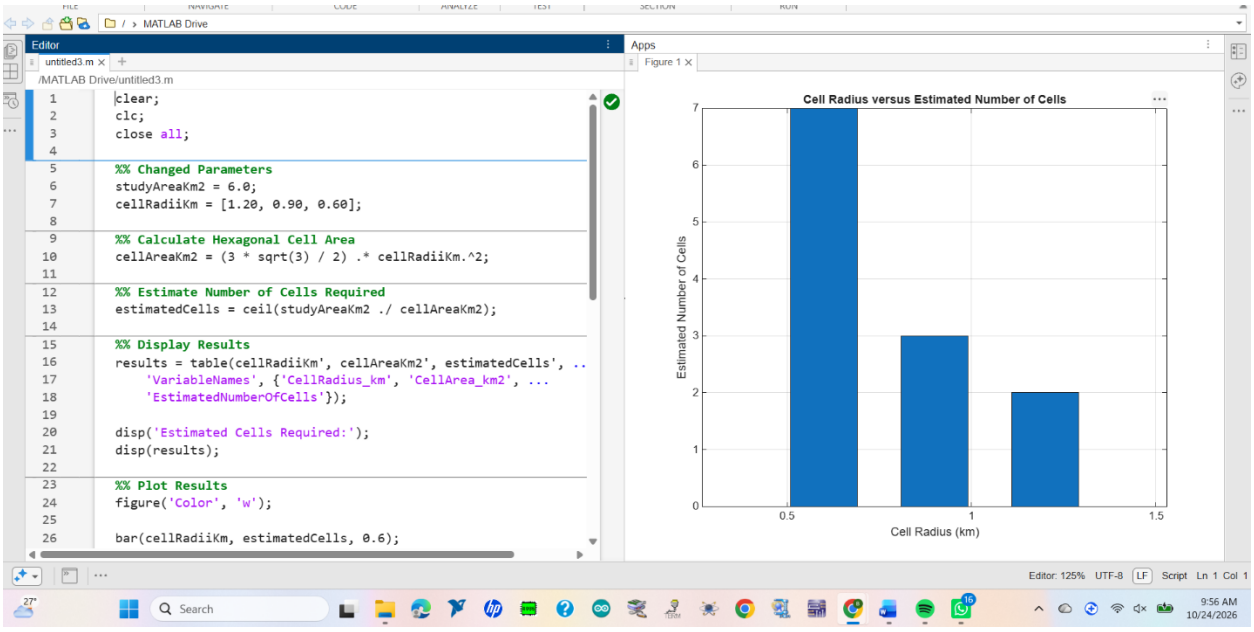
%% Estimate Number of Cells Required

estimatedCells = ceil(studyAreaKm2 ./ cellAreaKm2);

%% Display Results
```

```
results = table(cellRadiiKm', cellAreaKm2', estimatedCells', ...
    'VariableNames', {'CellRadius_km', 'CellArea_km2', ...
    'EstimatedNumberOfCells'});
disp('Estimated Cells Required:');
disp(results);
%% Plot Results
figure('Color', 'w');
bar(cellRadiiKm, estimatedCells, 0.6);
xlabel('Cell Radius (km)');
ylabel('Estimated Number of Cells');
title('Cell Radius versus Estimated Number of Cells');
grid on;
```

OUTPUT:



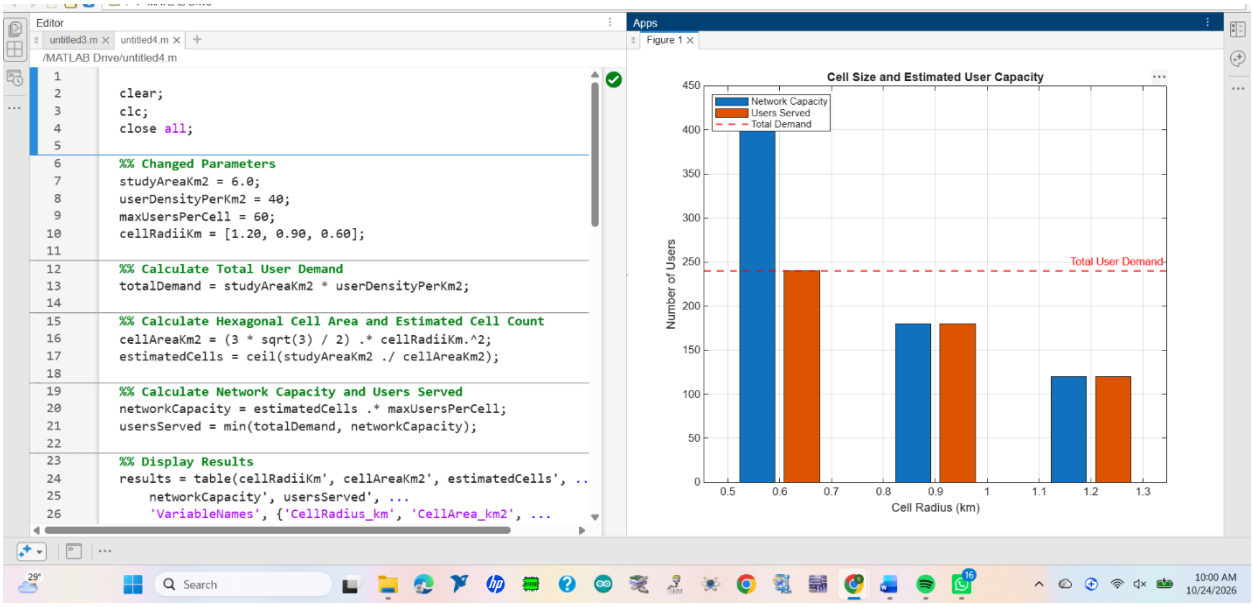
OBJ-3:CODE

```
clear;
clc;
close all;
%% Changed Parameters
studyAreaKm2 = 6.0;
userDensityPerKm2 = 40;
maxUsersPerCell = 60;
cellRadiiKm = [1.20, 0.90, 0.60];
%% Calculate Total User Demand
totalDemand = studyAreaKm2 * userDensityPerKm2;
%% Calculate Hexagonal Cell Area and Estimated Cell Count
cellAreaKm2 = (3 * sqrt(3) / 2) .* cellRadiiKm.^2;
estimatedCells = ceil(studyAreaKm2 ./ cellAreaKm2);
%% Calculate Network Capacity and Users Served
networkCapacity = estimatedCells .* maxUsersPerCell;
usersServed = min(totalDemand, networkCapacity);
%% Display Results
results = table(cellRadiiKm', cellAreaKm2', estimatedCells', ...
    networkCapacity', usersServed', ...
    'VariableNames', {'CellRadius_km', 'CellArea_km2', ...
    'EstimatedNumberOfCells', 'NetworkCapacity', 'UsersServed'});
disp('Estimated User Capacity:');
disp(results);
```

```
fprintf('Total User Demand: %d users\n', totalDemand);
%% Plot Results
figure('Color', 'w');
bar(cellRadiiKm, [networkCapacity', usersServed'], 'grouped');
hold on;
yline(totalDemand, '--r', 'Total User Demand', ...
    'LineWidth', 1.5);

xlabel('Cell Radius (km)');
ylabel('Number of Users');
title('Cell Size and Estimated User Capacity');
legend('Network Capacity', 'Users Served', 'Total Demand', ...
    'Location', 'northwest');
grid on;
hold off;
```

OUTPUT:

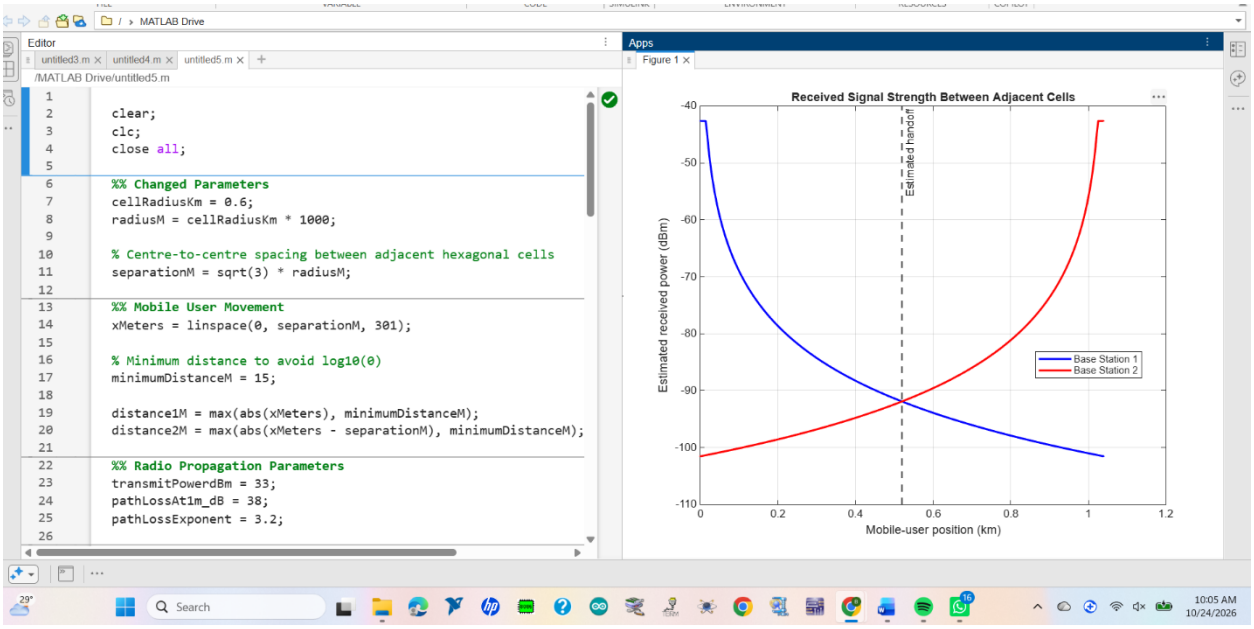


OBJ-4:CODE

```
clear;
clc;
close all;
%% Changed Parameters
cellRadiusKm = 0.6;
radiusM = cellRadiusKm * 1000;
% Centre-to-centre spacing between adjacent hexagonal cells
separationM = sqrt(3) * radiusM;
%% Mobile User Movement
xMeters = linspace(0, separationM, 301);
% Minimum distance to avoid log10(0)
minimumDistanceM = 15;
distance1M = max(abs(xMeters), minimumDistanceM);
distance2M = max(abs(xMeters - separationM), minimumDistanceM);
%% Radio Propagation Parameters
transmitPowerdBm = 33;
pathLossAt1m_dB = 38;
pathLossExponent = 3.2;
%% Calculate Received Signal Power
receivedPower1dBm = transmitPowerdBm - pathLossAt1m_dB ...
    - 10 * pathLossExponent .* log10(distance1M);
receivedPower2dBm = transmitPowerdBm - pathLossAt1m_dB ...
    - 10 * pathLossExponent .* log10(distance2M);
%% Determine Handoff Position
handoffIndex = find(receivedPower2dBm >= receivedPower1dBm, 1, 'first');
```

```
if isempty(handoffIndex)
    handoffPositionM = NaN;
    disp('Handoff position was not detected.');
else
    handoffPositionM = xMeters(handoffIndex);
    fprintf('Estimated handoff position: %.1f m from base station 1\n', ...
        handoffPositionM);
end
%% Plot Received Signal Strength
figure('Color', 'w');
plot(xMeters / 1000, receivedPower1dBm, ...
    'b', 'LineWidth', 1.8);
hold on;
plot(xMeters / 1000, receivedPower2dBm, ...
    'r', 'LineWidth', 1.8);
if ~isnan(handoffPositionM)
    xline(handoffPositionM / 1000, '--k', ...
        'Estimated handoff', 'LineWidth', 1.5);
end
xlabel('Mobile-user position (km)');
ylabel('Estimated received power (dBm)');
title('Received Signal Strength Between Adjacent Cells');
legend('Base Station 1', 'Base Station 2', ...
    'Location', 'best');
grid on;
box on;
hold off;
```

OUTPUT:


MATLAB Drive
Editor
untitled3.m
untitled4.m
untitled5.m
/MATLAB Drive/untitled5.m
clear;
clc;
close all;
%% Changed Parameters
cellRadiusKm = 0.6;
radiusM = cellRadiusKm * 1000;
% Centre-to-centre spacing between adjacent hexagonal cells
separationM = sqrt(3) * radiusM;
%% Mobile User Movement
xMeters = linspace(0, separationM, 301);
% Minimum distance to avoid log10(0)
minimumDistanceM = 15;
distance1M = max(abs(xMeters), minimumDistanceM);
distance2M = max(abs(xMeters - separationM), minimumDistanceM);
%% Radio Propagation Parameters
transmitPowerdBm = 33;
pathLossAt1m_dB = 38;
pathLossExponent = 3.2;
Apps
Figure 1
Received Signal Strength Between Adjacent Cells
Estimated handoff
Estimated received power (dBm)
Mobile-user position (km)
Base Station 1
Base Station 2
-40
-50
-60
-70
-80
-90
-100
-110
0
0.2
0.4
0.6
0.8
1
1.2
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